Thirty-Seven Years of Simulated Annealing on a Causal Set

A Revival of Bombelli (1987) with 2026 Tools

Jose Ignacio Martin Gandul · 2026

> "*An application of simulated annealing*" — Title of Appendix A.2, Luca Bombelli's PhD thesis, 1987

> "*¿Para qué llamar caminos*
> *a los surcos del azar?*"
> — Antonio Machado

Rafael Sorkin and Luca Bombelli introduced causal set theory in 1987: the idea that spacetime, at the Planck scale, may be described by a locally finite partially ordered set. The same year, Bombelli appended to his thesis a Pascal program that tried to embed small causal sets into Minkowski spacetime by simulated annealing. The program ran on the workstations of the era, produced results for a handful of cases, and was never published as a standalone tool.

This note documents a small experiment: port Bombelli's program to Python, run it reproducibly on ordinary machines, and compare many runs instead of one. It is meant as a readable revival for people who like physics and computation, not as a formal claim of advancing causal set theory.

I. The Instruments

Aspect	Bombelli (1987)	This work (2024–2026)
Language	Pascal	Python 3.12 (faithful port)
Hardware	~1 MIPS workstation	Ordinary CPU machine
Runs per case	Single run	Ensemble of K seeds (K ≥ 8)
Schedule selection	Fixed by hand	Empirical grid scan
Warmup protocol	10 unconditional accepts	Three modes compared: legacy / skip / guarded
Non-manifoldlike controls	None	Kleitman–Rothschild orders, suspended corona posets
Dimension estimators	None (annealing only)	Myrheim–Meyer, Meyer midpoint scaling
Structural invariants	None	15 order-theoretic features (chain counts, link density, height, ...)
Correlation analysis	None	Spearman ρ, partial Spearman controlling for *n*

Aspect	Bombelli (1987)	This work (2024–2026)
Reproducibility	Thesis appendix	Frozen benchmark CSV + Python scripts in this repository

II. What Becomes Easier To Check

The original thesis demonstrated the method on very small causets where a single run could be completed in minutes on the hardware of the day. The table below compares that single-run style with small CPU ensembles. All 2026 entries use the same energy function and move set as the original.

Causal set size *n*	1987-style run	2026 CPU ensemble	Note
6	Single run, qualitative	Success rate 25–50 %, small grid available	Best cells: *dim* = 3–4
12	Single run, qualitative	Success rate 0–12.5 %, schedule matters strongly	Benchmark `tesis_like_12.in`
16	Not reported	Success rate 0–12.5 %, budget-dependent	Small runs still possible
24	Not accessible	All runs time out at default budget	Too slow for this simple setup
32–64	Not accessible	Ensemble statistics available; floor pathology characterised	Phases 4A–4D

"*Not accessible*" here means only that this simple historical method does not scale comfortably under the default budget.

III. The Same Benchmark, Thirty-Seven Years Apart

The reproducible input `tesis_like_12.in` — twelve elements, a tesis-like causal structure — serves as the common witness.

Metric	Bombelli defaults (T_0 = 100, α = 0.9)	Tuned schedule (T_0 = 180, α = 0.8)	Change
Mean final energy (100 seeds)	20.021	0.166	−99.2 %
Zero-energy runs	0 / 100	0 / 100	—
Schedule selected by	Intuition / thesis norm	Empirical grid scan	—

The tuned schedule uses the same annealer, the same energy function, and the same move set as the original. The only difference is two numbers. This is not a flaw in the 1987 work; it is just a reminder that annealing schedules can matter a lot, and repeated runs make that easier to see.

IV. What the Warmup Was Doing

One set of runs looked at the warmup phase. The original warmup phase makes 10 unconditional accept steps before annealing begins — a design choice intended to equilibrate the system at high temperature. The table below shows what those 10 steps do to controlled initializations.

Initialization	Legacy warmup	Skip warmup	Guarded warmup
Ground truth (E = 0)	Preserved 18/18, E = 0.000	Preserved 18/18, E = 0.000	Preserved 18/18, E = 0.000
Truth + small noise ($\varepsilon = 10^{-3}$)	Mean E = 18.92, 16/18	Mean E = 12.12, 17/18	Mean E = 0.001, **18/18**
Truth + medium noise ($\varepsilon = 5\times10^{-2}$)	Mean E = 395.8, 0/18	Mean E = 286.0, 0/18	Mean E = 255.3, 0/18
Random initialization	Mean E = 405.2, 8/18	Mean E = 307.9, 11/18	Mean E = 271.4, **12/18**

Grid: $d \in \{2, 3, 4\}$, $n \in \{32, 64\}$, seeds 1959/1962/1987 — 18 cells per row.

The guarded warmup accepts a proposed move only if it does not increase the energy (greedy descent). It is an external wrapper around the same `ConesSimulator` internals — no change to the energy, the move set, or the cooling schedule. In the small-noise cases tested here, this simple guard preserves near-truth configurations much better.

The oracle check (Phase 2C) confirmed that the energy formula returns exactly 0.0 at ground-truth coordinates in 18/18 cases. The Bombelli energy behaves as expected on those controlled inputs; the large errors above come from the search dynamics rather than from the energy formula itself.

V. What the Order-Theoretic Invariants Reveal

The 1987 program had no way to ask "is this causal set manifoldlike before we try to embed it?" Phase 1 of this study adds that question.

The table below compares two independent dimension estimators — the Myrheim–Meyer formula and Meyer's midpoint scaling — on Minkowski sprinklings (manifoldlike by construction) and two non-manifoldlike controls (Kleitman–Rothschild three-layer orders and suspended corona posets), at n = 256, ensemble of 5 seeds.

Family	d_spacetime	n	MM dim	midpoint dim	|discrepancy|
Minkowski	2	256	**2.02**	**2.06**	0.07
Minkowski	3	256	**3.06**	**3.00**	0.28
Minkowski	4	256	**4.07**	**3.80**	0.42
Kleitman–Rothschild	—	256	2.37	4.71	**2.34**
Corona poset	—	256	1.98	7.00	**5.02**

For manifoldlike sprinklings the two estimators stay close to the expected dimension. For non-manifoldlike controls they separate. The finite-size trend is a useful warning sign: the causal set may not be well described by a low-dimensional sprinkling.

VI. What the Correlate Audit Found

Phase 4D (the robustness-vs-invariants audit) asked whether any order-theoretic property of a causal set predicts how unstable the embedding optimizer is across different random seeds. At per-seed level (N = 90 observations, n ∈ {32, 48, 64}):

Invariant	Target robustness metric	Raw Spearman ρ	Partial ρ (controlling for *n*)
`relation_count`	Floor saturation fraction	+0.941	+0.903
`chain2_count`	Floor saturation fraction	+0.941	+0.903
`height`	Floor saturation fraction	+0.836	+0.779
`abs_discrepancy_mm_midpoint`	Floor saturation fraction	−0.759	−0.659
`mm_dim`	Floor saturation fraction	−0.698	−0.530

The correlation survives controlling for finite-size scaling: even within a fixed *n* stratum, denser, taller causal sets are harder for the optimizer in these runs. This is a statement about this optimizer's difficulty landscape, not about physical embeddability.

VII. A Timeline

Year	Event
1987	Sorkin and Bombelli introduce causal set theory (*Phys. Rev. Lett.* 59, 521)
1987	Bombelli writes the Pascal annealing program (PhD thesis, Appendix A.2)
1987–2024	Program dormant. CST develops theoretically. Modern tools emerge.
2024	Faithful port to Python 3.12, validated against thesis inputs
2024	Faithful CPU implementation made portable and reproducible
2024–2025	Phases 1–3: structural atlas, embedding bridge, schedule probe, warmup audit
2025–2026	Phases 4–5: epsilon sweep, survival probe, seed robustness, morphology audit
2026	Phase 4D n-control extension: correlates survive finite-size correction

Year	Event
2026	This note

VIII. What Remains the Same

The energy function is Bombelli's. The move set is Bombelli's. The cooling rule (`4 × exp(−ΔE / T)`) is Bombelli's. The input format is Bombelli's. The core loop is Bombelli's.

Everything else — the number of runs, the parameter scan, the controls, the invariants, the n-control audit — is a modern way to look more carefully at the same small program.

Acknowledgements

This work is a revival and empirical characterisation of the computational program introduced in:

> L. Bombelli, *Space-time as a Causal Net*, PhD thesis, Syracuse University, 1987.

and motivated by the foundational paper:

> L. Bombelli, J. Lee, D. Meyer, R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521 (1987).

The code in this repository is intentionally portable so it can run on ordinary machines.

Old code, rerun carefully, can still teach us something modest.